Lesson 9-1 · Session 1

Independent and Dependent Variables

Name _____ Date _____ Period _____

Maryland Standard 6.AT.11 · Learning target — I can name the independent and dependent variables in a relationship.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

Learn from a model

REMEMBER

- The independent variable is the input (often x). The situation limits which input values make sense.
- The dependent variable is the output — its value DEPENDS on the independent variable (often y).
- Watch out — time is usually independent, but always check what the situation says.
- Check it — say the sentence "___ depends on ___" and see whether it makes sense.

WORKED EXAMPLE

Naming the Independent and Dependent Variables Jaden walks dogs after school. The more hours he walks, the more money he earns. Which variable is independent, and which is dependent?

Step 1 Name the two quantities: hours walked and money earned.

Step 2 Which is chosen? Jaden picks his hours — hours walked is the independent variable.

Step 3 Which responds? The money changes because the hours change — money earned is dependent.

Step 4 Sentence check: 'Money earned depends on hours walked.' The sentence makes sense.

Answer Independent variable: hours walked. Dependent variable: money earned.

Practice

1. In the situation 'the total cost depends on the number of items bought,' which is the independent variable? _____ Which is the dependent? _____

Answer _____

RECORD YOUR RESPONSE

2. In 'distance traveled depends on the number of hours driven,' identify the independent and dependent variables.

Answer _____

RECORD YOUR RESPONSE

3. Write your own real-world situation and identify the independent and dependent variables.

Answer _____

RECORD YOUR RESPONSE

4. In each pair of quantities, write I next to the independent variable and D next to the dependent variable.

a. hours of basketball practice _____ free throws made _____
b. bags of soil bought _____ total price paid _____
c. days a seedling grows _____ height of the seedling _____
d. guests at a party _____ slices of pizza needed _____
e. minutes on a treadmill _____ Calories burned _____

Answer _____

RECORD YOUR RESPONSE

5. Swim Practice. Kalinda decides how many laps to swim at the rec center pool. The more laps she swims, the more minutes she stays in the water. Name the independent variable and the dependent variable.

Answer _____

RECORD YOUR RESPONSE

6. Snow Shoveling. Deshawn clears snow from sidewalks on his block. He charges the same amount for every sidewalk he clears. Write a sentence that uses 'depends on' to connect the two quantities, then label each variable.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

7. School Library. Ms. Okafor keeps track of the school library. On days when the library stays open longer, more students sign in. She also sees that when more students sign in, more books get checked out. Name the independent and dependent variable in each of the two pairs.

Answer _____

RECORD YOUR RESPONSE

8. Tomás reads this situation: ‘The oven temperature a baker sets changes how long a cake needs to bake.’ He says the baking time is the independent variable. Explain whether Tomás is right. Support your answer with a sentence that uses ‘depends on.’

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

9. A car’s fuel use depends on the miles driven. Name the independent and dependent variables.

Answer _____ both variables

RECORD YOUR RESPONSE

10. A plant’s height depends on the days since planting. Name both variables.

Answer _____ both variables

SHOW YOUR THINKING

11. Which Axis? A graph shows total pay against hours worked. Name the variable on each axis.

Answer _____ both axes

RECORD YOUR RESPONSE

12. Find the Mistake. Kim says the total cost is independent because it is the bigger number. Explain the mistake.

Answer _____ the mistake

EXPLAIN WITH MATHEMATICAL EVIDENCE

Spiral review

13. From Lesson 8-5 Session 1. Describe the graph of $n < 5$.

Answer _____

RECORD YOUR RESPONSE

14. From Lesson 8-2 Session 2. Solve $m - 15 = 22$.

Answer _____

SHOW YOUR THINKING

15. From Lesson 6-8 Session 2. Factor $12n + 20$.

Answer _____

SHOW YOUR THINKING

Test practice

16. The dependent variable goes on the —

□ A. origin	□ B. x-axis
□ C. y-axis	□ D. scale

17. In “minutes of exercise and calories burned,” the dependent variable is —

- □ A. the person
- □ B. minutes of exercise
- □ C. the exercise
- □ D. calories burned

18. The variable you control is the —

- □ A. independent variable
- □ B. dependent variable
- □ C. constant
- □ D. coefficient

Check and revise

CHECK YOUR MATHEMATICS	HOW DID IT GO?
Check every pair using the same rule and label quantities and units. Every part answered / Work shown / Units labeled when needed	I can do this: Independently / With a hint / With help Question to revisit: _____

One correction or question I want to discuss: _____

Substitute one input-output pair into your rule. Check units and whether your answer fits the situation.